

## IWAKUNI MCAS, Japan

WMO: 477640

Lat: 34.145N

Lon: 132.247E

Elev: 20

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43324

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 30.3       | 32.4 | 17.2                          | 13.1 | 38.1 | 19.3 | 14.5 | 38.4 | 22.4                  | 42.6 | 20.3 | 43.5 | 6.4                   | 300  | 0.400 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 10.9                   | 91.3            | 79.5 | 89.2 | 78.5 | 87.5 | 78.2 | 81.0                | 87.7 | 79.9 | 86.6 | 78.9 | 85.7 | 9.1                  | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.0                            | 150.4 | 85.6 | 77.8 | 144.7 | 84.9 | 76.8 | 139.8 | 84.2 | 44.8          | 88.1 | 43.6 | 86.6 | 42.4 | 85.9 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 20.3              | 17.4 | 15.2 |                            | DB   | 27.3 | 94.5               | 2.6                                                | 1.8       | 25.4 | 95.9       | 23.8 | 96.9       | 22.3 | 98.0       | 20.4 | 99.3 |
|                   |      |      |                            | WB   | 24.4 | 83.1               | 2.3                                                | 1.7       | 22.7 | 84.3       | 21.4 | 85.3       | 20.1 | 86.2       | 18.4 | 87.4 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 61.6   | 42.5  | 43.1  | 48.9  | 57.8  | 66.3  | 72.5  | 80.1  | 82.3  | 75.7  | 66.2  | 55.9  | 46.4 |
|                                                                               | DBStd       | 14.49  | 4.14  | 4.48  | 5.27  | 5.39  | 3.88  | 3.76  | 4.14  | 3.14  | 4.29  | 4.95  | 5.54  | 4.97 |
|                                                                               | HDD50       | 670    | 234   | 198   | 86    | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 11    | 134  |
|                                                                               | HDD65       | 2959   | 697   | 613   | 500   | 222   | 31    | 1     | 0     | 0     | 0     | 43    | 275   | 577  |
|                                                                               | CDD50       | 4891   | 2     | 4     | 51    | 239   | 504   | 676   | 932   | 1000  | 770   | 502   | 189   | 22   |
|                                                                               | CDD65       | 1707   | 0     | 0     | 0     | 5     | 70    | 227   | 467   | 535   | 320   | 80    | 3     | 0    |
|                                                                               | CDH74       | 14113  | 0     | 0     | 0     | 12    | 177   | 911   | 4483  | 5985  | 2306  | 238   | 1     | 0    |
|                                                                               | CDH80       | 4262   | 0     | 0     | 0     | 0     | 10    | 112   | 1450  | 2205  | 472   | 13    | 0     | 0    |
| Wind                                                                          | WSAvg       | 6.5    | 7.0   | 6.9   | 6.9   | 6.2   | 5.8   | 5.2   | 5.8   | 6.2   | 6.6   | 7.1   | 6.7   | 7.1  |
| Precipitation                                                                 | PrecAvg     | 66.6   | 2.5   | 2.9   | 5.3   | 7.3   | 7.3   | 11.5  | 8.9   | 5.0   | 7.1   | 4.3   | 3.2   | 1.7  |
|                                                                               | PrecMax     | 102.5  | 6.7   | 8.1   | 11.5  | 13.9  | 14.8  | 20.5  | 27.4  | 18.6  | 15.4  | 11.3  | 8.0   | 4.1  |
|                                                                               | PrecMin     | 44.5   | 0.2   | 0.4   | 0.5   | 2.2   | 2.1   | 2.3   | 0.2   | 0.1   | 0.2   | 1.0   | 0.1   | 0.1  |
|                                                                               | PrecStd     | 12.5   | 1.6   | 1.7   | 2.7   | 2.5   | 3.2   | 4.5   | 5.6   | 3.5   | 3.5   | 2.6   | 1.9   | 1.1  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 57.3  | 60.1  | 67.3  | 75.5  | 80.9  | 86.3  | 93.3  | 93.7  | 89.3  | 82.3  | 71.8  | 62.7 |
|                                                                               |             | MCWB   | 48.9  | 52.8  | 55.7  | 61.8  | 66.8  | 75.6  | 79.4  | 80.1  | 77.6  | 73.0  | 63.3  | 55.3 |
|                                                                               | 2%          | DB     | 53.8  | 55.7  | 64.1  | 71.3  | 78.2  | 82.8  | 91.1  | 91.7  | 86.3  | 78.9  | 69.7  | 59.4 |
|                                                                               |             | MCWB   | 46.8  | 48.9  | 55.3  | 60.2  | 65.9  | 73.7  | 79.1  | 79.9  | 76.7  | 69.9  | 61.9  | 52.0 |
|                                                                               | 5%          | DB     | 51.9  | 53.4  | 60.9  | 68.6  | 75.6  | 80.6  | 88.5  | 89.7  | 84.3  | 76.7  | 67.5  | 57.1 |
|                                                                               |             | MCWB   | 44.9  | 46.7  | 53.4  | 59.4  | 65.1  | 72.3  | 78.6  | 78.8  | 75.3  | 67.6  | 60.5  | 50.7 |
|                                                                               | 10%         | DB     | 50.0  | 51.4  | 57.6  | 66.3  | 73.7  | 78.8  | 86.6  | 88.0  | 82.6  | 74.8  | 65.1  | 55.2 |
|                                                                               |             | MCWB   | 43.9  | 45.0  | 51.1  | 58.4  | 64.3  | 71.2  | 78.1  | 78.4  | 74.1  | 66.1  | 58.5  | 48.9 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 52.2  | 54.6  | 59.8  | 65.7  | 71.1  | 77.1  | 82.3  | 82.8  | 80.1  | 74.7  | 67.3  | 57.8 |
|                                                                               |             | MCDB   | 55.1  | 57.3  | 63.6  | 71.2  | 76.1  | 83.2  | 89.0  | 90.1  | 86.2  | 79.7  | 70.1  | 61.0 |
|                                                                               | 2%          | WB     | 48.7  | 50.6  | 56.8  | 63.4  | 69.1  | 75.5  | 80.8  | 81.4  | 78.2  | 72.5  | 64.5  | 53.9 |
|                                                                               |             | MCDB   | 52.1  | 53.9  | 61.2  | 67.8  | 74.5  | 80.8  | 87.3  | 88.5  | 84.3  | 77.1  | 67.9  | 57.6 |
|                                                                               | 5%          | WB     | 46.2  | 48.2  | 54.6  | 61.6  | 67.5  | 74.2  | 79.8  | 80.3  | 77.0  | 69.6  | 61.9  | 51.3 |
|                                                                               |             | MCDB   | 50.3  | 51.8  | 59.7  | 66.4  | 73.4  | 78.9  | 86.4  | 87.2  | 82.5  | 74.7  | 65.8  | 55.9 |
|                                                                               | 10%         | WB     | 44.1  | 45.9  | 51.9  | 59.8  | 65.9  | 72.7  | 78.7  | 79.3  | 75.8  | 67.2  | 59.4  | 49.5 |
|                                                                               |             | MCDB   | 49.1  | 50.0  | 57.2  | 65.0  | 71.8  | 77.1  | 85.4  | 86.1  | 81.2  | 73.0  | 64.3  | 54.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 12.9  | 13.1  | 14.2  | 14.2  | 13.7  | 10.9  | 10.3  | 10.9  | 11.7  | 13.4  | 13.8  | 13.1 |
|                                                                               |             | MCDBR  | 15.0  | 15.3  | 17.3  | 17.4  | 16.7  | 13.3  | 12.8  | 13.0  | 12.5  | 14.3  | 14.5  | 14.9 |
|                                                                               |             | MCWBR  | 10.5  | 11.5  | 12.2  | 11.0  | 9.9   | 7.3   | 6.5   | 6.6   | 6.7   | 8.5   | 9.9   | 10.6 |
|                                                                               | 5% WB       | MCDBR  | 13.9  | 13.3  | 15.3  | 13.7  | 13.5  | 11.1  | 11.8  | 11.7  | 10.7  | 11.6  | 12.7  | 13.3 |
|                                                                               |             | MCWBR  | 10.9  | 11.3  | 12.3  | 10.0  | 9.4   | 6.8   | 6.6   | 6.7   | 6.3   | 8.2   | 10.2  | 10.4 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.406  | 0.448 | 0.510 | 0.539 | 0.540 | 0.557 | 0.557 | 0.527 | 0.487 | 0.441 | 0.418 | 0.400 |      |
|                                                                               | taud        | 2.141  | 2.051 | 1.903 | 1.886 | 1.949 | 1.989 | 2.037 | 2.126 | 2.196 | 2.228 | 2.200 | 2.173 |      |
|                                                                               | Ebn at Noon | 245    | 247   | 242   | 242   | 244   | 239   | 238   | 243   | 247   | 248   | 240   | 239   |      |
|                                                                               | Edh at Noon | 38     | 47    | 59    | 62    | 59    | 57    | 54    | 49    | 43    | 39    | 36    | 36    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 695    | 912   | 1244  | 1525  | 1680  | 1453  | 1559  | 1618  | 1296  | 1078  | 791   | 617   |      |
|                                                                               | RadStd      | 56     | 76    | 96    | 138   | 172   | 156   | 213   | 161   | 129   | 107   | 65    | 43    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | +0.67 | N/A     | +1.84  | N/A     | N/A   | N/A   | N/A         | N/A   | +203  | N/A   |
| (47) | Regional (32 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page